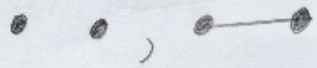
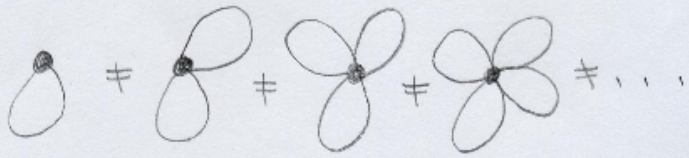
1. How many isomorphic graphs of order 1 are there? of order 2? of order 3? Explain why the answer to each of the preceding question is ∞ for general graphs.

order 1 ..... one !

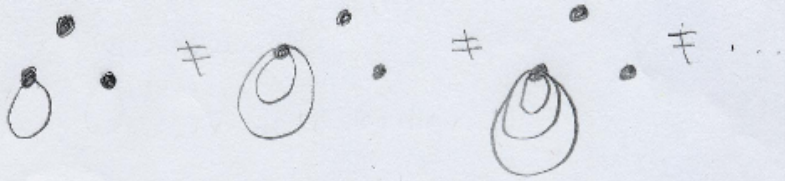
order 2  .... two !

order 3 ..... four !

For general graphs,

order 1 

order 2

order 3 

$\therefore$ The number of non-isomorphic graphs of order 1, order 2, order 3 is ∞.

2. Does there exist a graph of order 5 whose degree sequence equal (4,4,3,2,2)

Sol.) Answer : No.

By theorem, sum of degree sequence (4,4,3,2,2) has to be even

but sum of degree of given sequence (4,4,3,2,2) is odd.

5. Let G is a graph of order n $(n \geq 2)$, and $v_1, v_2, \dots, v_n$ are vertices of G. We can make a degree of some vertex v_i $(1 \leq i \leq n)$ at least 0, at most $n-1$. However $n-1$ degree and 0 degree can't occur at the same time. So we can make degrees of $0 \sim (n-2)$ or $1 \sim (n-1)$. So, by PHP, some two of n vertices has same degree.

$$\begin{array}{ll} & \text{Vertices} \quad \text{degree} \\ n\left\{\begin{array}{l} v_1 \\ v_2 \\ \vdots \\ v_n \end{array}\right. & \left.\begin{array}{l} 0 \\ 1 \\ \vdots \\ (n-2) \end{array}\right\} n-1 \end{array} \quad \text{or} \quad \begin{array}{ll} & \text{vertices} \quad \text{degree} \\ n\left\{\begin{array}{l} v_1 \\ v_2 \\ \vdots \\ v_n \end{array}\right. & \left.\begin{array}{l} 0 \\ 1 \\ \vdots \\ (n-2) \end{array}\right\} n-1 \end{array}$$

Most people missed this point. It is important.

20) Let G be a disconnected graph on n vertices. If G has a component on m vertices, then none of the $m(n-m)$ possible edges between this component and the rest of G are in G.

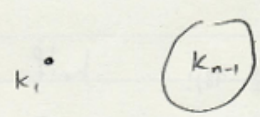
Now $f(m) = m(n-m) = mn - m^2$, so $f''(n) = -2$, so f is minimised on the set $M = \{1, \dots, n-1\}$ at 1 and $n-1$. That is $m(n-m) \leq f(1) = n-1$.

So if G is disconnected it has at most

$$\binom{n}{2} - (n-1) = \frac{n(n-1)}{2} - (n-1) = \frac{(n-2)(n-1)}{2}$$

edges. This implies that any graph with $\frac{(n-2)(n-1)}{2} + 1$ edges is connected.

The graph $G = K_1 \cup K_{n-1}$:



has exactly $\binom{n}{2} - (n-1) = \frac{(n-2)(n-1)}{2}$ edges.

13) Assume vertices u and v are joined by a walk in G. Let $u = v_1, v_2, \ldots v_n = v$ be the shortest such walk in G. If this isn't a path, then there exist $i < j$ such that $v_i = v_j$. But then

$$v_1, v_2 \ldots v_i, v_{j+1}, \ldots, v_n$$

is a shorter walk from u to v, contradicting our assumption above. So $v_1, v_2, \ldots v_n$ is a path from u to v.

54) A path has an eulerian path. Any other tree has a vertex of degree 3 or more so Any eulerian walk would have to visit this vertex at least twice, so wouldn't be a path. Thus only paths have eulerian paths.

(One can actually show the only trees with eulerian walks are paths. This is maybe what they meant to ask.).

66) A connected graph with n vertices and n edges has exactly one cycle.

Pf Let G be a connected graph with n vertices and n edges. There is a spanning subtree T with $n-1$ edges. Let uv be the edge in G but not T. Any cycle in G must contain uv, or else it would be in T also, and for any cycle $uv_1v_2 \ldots v_rvu$ in G, $uv_1 \ldots v_rv$ is a path from u to v in T. As T is a tree, there is a unique such path. So G has a unique cycle.